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# A Randomized Trial of Healthy Weight Gain in Athletic Individuals

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**ABSTRACT**

Many athletes desire weight gain, ideally as lean body mass (LBM), to improve athletic performance. These athletes are commonly encouraged to increase energy intake by ~500 kcal/day with emphasis on adequate protein, carbohydrate, and judicious inclusion of healthy fat-containing energy-dense foods, such as peanuts/peanut butter, along with rigorous resistance training (RT). These guidelines target gains of ~0.23 kg/week. However, little is known about the efficacy of such regimens. **Purpose:** To evaluate the outcomes of a 10-week diet and exercise regimen designed to promote healthy weight gain with excess energy from peanut-containing or high-carbohydrate foods. **Methods:** 19 male and 13 female athletes were randomly assigned to receive 500 additional kcal/day above typical intake through provision of either peanut-based whole foods/snacks (PNT group) or a similar, high-carbohydrate, peanut-free snack (CHO group) along with supervised, whole-body RT (3 days/week for 60-120 minutes). Body composition was assessed by dual-energy x-ray absorptiometry at baseline and post-intervention. Results: Total body mass (TBM) increased 2.2±1.3 kg with 1.5±1.1 kg as LBM after week 10. The PNT group (N=16; 27±7 years; 10 men, 6 women) gained less TBM than the CHO group (N=16; 23±3 years; 9 men, 7 women) (1.6±1.1 kg vs 2.7±1.2 kg, respectively, $P$=0.007) with no differences in LBM (1.2±1.1 kg vs 1.9±1.0 kg, $P$=0.136). **Conclusions:** These results suggest that the addition of 500 kcal/day from whole foods/snacks in combination with a rigorous RT program promotes a similar weight gain of ~0.22 kg/week, primarily as LBM, over 10 weeks in both male and female athletes. However, snack macronutrient content may impact the effectiveness of this regimen.

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**Key Words**: ENERGY SURPLUS, HYPERENERGETIC INTAKE, RESISTANCE TRAINING, POSITIVE ENERGY BALANCE, MUSCLE HYPERTROPHY, OVERFEEDING

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## INTRODUCTION

A variety of athletes intentionally attempt weight gain to improve strength and power for athletic performance and to equal opponents' size in contact sports (1). Current sports nutrition recommendations to promote healthy weight gain (0.23 to 0.45 kg/week; wk) include increasing energy intake by ~500 kcal/day (1-6), consuming protein within the range of 1.2 to 2.0 g protein per kg body weight/day, adequate carbohydrate (CHO) (7, 8), and healthy fats from foods such as peanuts, peanut butter, nuts, and avocado, in combination with rigorous resistance training (RT) (7, 9). Surprisingly little research has been conducted to either help establish or validate the typical aforementioned textbook guidelines (10), which were initially established using estimates of the energy required to promote muscle tissue accretion (7) from studies conducted in malnourished (11) and experimentally overfed sedentary subjects (12). Moreover, the additional 500 kcal/day was established based on estimates of the National Research Council that 5 to 5.65 kcal are needed to support accretion of 1 gram of tissue during growth (7). However, there is a lack of evidence to support which macronutrient or combination of macronutrients should comprise the energy surplus for healthy weight gain (10).

In the mid-to-late 1990s, weight gain in athletes was a hot topic (8, 9), yet the focus seemed to divert to protein intake and timing on acute muscle protein synthesis (MPS) (13). A recent review on weight gain in athletes found only seven studies (14-20) have been conducted over the last 35 years, with only two conducted since 2016 (18, 20). Bartels et al. (14) demonstrated approximately 0.3 kg increase in lean body mass (LBM)/week and no increase in body fat percent in 12 experienced male weight trainers with the addition of 500 extra kcal/day during nine weeks of heavy RT. Body weight gains did not differ whether the hyperenergetic diet

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was 40% versus 65% CHO. A recent study in 20 resistance-trained men who consumed ≥647.5 kcal/day from a protein and CHO mass gainer supplement along with a 3-day/wk RT program for six weeks demonstrated total body mass (TBM) and LBM gains of 3.6±1.7 kg and 2.4±1.6 kg, respectively (21). Furthermore, to our knowledge, only two previous studies (16, 19) evaluating intentional weight gain included a small number of active women, who may also desire weight gain (22) and should be equally represented in future studies.

Research suggests that active individuals interested in weight gain often rely on commercially-available powders (15), energy bars, and other high-protein specialized products as a source of additional calories (22). These food items, however, are considered ultra-processed foods (UPF), which may be linked to adverse health outcomes, including increased risk of cardio-metabolic disease, stroke, and depression (23). Given the increased interest in the health benefits of whole-food and plant-based diets (and the combination of the two), peanuts may be one potential source of high-quality plant-based protein for athletes interested in weight/lean mass gain. Additionally, peanuts have a long shelf-life, are affordable, and are well-liked by most individuals (24).

Therefore, this study was conducted to address knowledge gaps related to healthy weight gain in athletes and to design better weight gain meal plans. The aim of the study was to determine the efficacy of the standard hyperenergetic sports performance diet (500 kcal surplus per day from adequate protein, CHO, and fat) and RT regimen with excess calories from either peanut/peanut-containing (PNT) or high-CHO (CHO) snack foods on weight and lean mass gains over a 10-wk period in athletic men and women desiring weight gain. We hypothesized that the

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standard hyperenergetic sports performance diet and RT regimen with excess calories from peanut or high-CHO snack foods would result in an average weight gain of 0.23 kg/wk equally in both snack groups and in both men and women, with the majority gained as LBM. Within this framework, a secondary aim was to characterize the health outcomes with the standard hyperenergetic and RT regimen.

## METHODS

### Participants

We aimed to recruit 32 healthy athletic men and women (18-40 years of age) who expressed the desire to gain body weight to enhance effectiveness in sport, fitness, or military training/competition. This convenience sample was based on financial and time constraints. To be eligible, participants had to be in good general health, be currently involved in sport, military, or training at the recreational, collegiate, or elite level, and had to have participated in RT as part of athletic or military training within the past year.

Participants were excluded if they had medical and/or orthopedic limitations that might affect the ability to fully participate in the study, current consumption of any dietary supplements or prescribed pharmacological agents that might affect lean tissue accretion, tobacco use, coronary heart disease, hypertension (and/or on blood pressure (BP) medication), history of or current signs of disordered eating (score >20 on the Eating Attitudes Test; EAT-26), current or past history of anabolic steroid use, weekly workout volume that greatly exceeded our RT protocol, an allergy to peanuts and/or tree nuts, or were unwilling to consume all foods/beverages provided. Participants were also excluded after screening if they had abnormal

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thyroid stimulating hormone (TSH) or hemoglobin (Hgb) concentrations that were outside of the reference ranges (TSH 0.45-4.50 uIU/ml and Hgb 13.0-17.7 g/dL for male or 11.1-15.9 g/dL for female participants). If participants were taking creatine, they were requested to stop use for a minimum of 30 days before starting the intervention. Women were excluded if they were pregnant, breastfeeding, amenorrhoeic or oligomenorrheic (which may be a sign of undereating).

Participants were recruited through flyers placed at strategic locations (college and community gyms and fitness and recreational centers) and emailed to athletic clubs and targeted listservs, and through partnering with the university's Corps of Cadets. The study was approved by the Institutional Review Board (IRB) for Human Subjects at Virginia Tech (VT, #NCT05426109), and written informed consent was obtained from all participants. The full protocol can be accessed at the clinicaltrials.gov site or at VT IRB (#21-561).

**Experimental Design**

Participants completed a 10-wk diet and RT regimen designed to promote healthy weight gain (0.23-0.45 kg/wk). Participants were screened and block randomized to the peanut (PNT) or non-peanut (carbohydrate; CHO) group based on sex (male or female) and RT experience (high or low), which was used to include comparable numbers of each in the two groups. For 10 weeks, both PNT and CHO groups received an additional 500 kcal per day from snacks and participated in three supervised RT sessions per week. Body composition (including LBM, fat mass (FM), and body fat percent), diet, activity level, fasting blood glucose and serum lipids, resting BP and heart rate (HR), muscular strength, aerobic fitness ($VO_2$max), and circumferences were assessed at baseline (BSL) and post-intervention with some measurements also assessed

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during the intervention at weeks 3 and 7. Figure 1 provides an overview of the study design and procedures.

**Preparticipation Screening**

Prior to enrollment, detailed diet and exercise data were collected and assessed by a Registered Dietitian (RD; AS) to ensure participants were consuming adequate energy (relative to estimated energy expenditure) and sufficient protein (>1.2 g/kg) from a balanced diet at BSL, and that their desire to gain weight was not due to wanting to participate in a research study or due to general undereating, unhealthy eating patterns, or disordered eating assessed by EAT-26.

RT background was assessed by asking detailed questions about training history. Participants were also asked about their best 1-repetition (rep) maximum (max) barbell back squat and bench press and their corresponding body weight when their 1-rep max was achieved. If the best 1-rep max was unknown, the amount of weight that could be lifted for 10 reps was requested. RT history was confirmed by types of lifts and number of reps and sets performed weekly over the previous year. This information was then assessed by one of the study's investigators with extensive RT experience (GH) who assigned individuals to the high or low RT experience groups. Other physical activity participation was assessed by the Baecke physical activity questionnaire (25) validated to assess and categorize physical activity into sport, work, and leisure-time activities.

A study investigator (EM) generated the random allocation sequence (Research Randomizer, www.randomizer. org). Another study investigator (AS) enrolled participants and

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used the randomization scheme generated to assign them to the intervention.

## Weight Gain Intervention

***Diet.*** The PNT group was provided with a variety of minimally processed whole peanut and peanut butter-containing snacks prepared in our metabolic kitchen. The snacks were calculated to contain 500+/- 10 kcal and 20 grams of protein using Nutrition Data System for Research nutrient analysis software (Version 2020, Nutrition Coordinating Center, Minneapolis, MN, USA). The CHO group was provided similar 500 kcal high CHO- containing snacks that contained ~5 grams of protein. The macronutrient profile of the snacks provided to the PNT group was 14%, 31%, and 55%, and to the CHO group was 3.5%, 63%, and 34% for protein, CHO, and fat, respectively. According to the NOVA classification system (26), the average UPF% by kcal for the PNT and CHO snacks was 16% and 30%, respectively.

Participants in both groups received two of the 10 possible snack options each week (three of one option, four of the other) (Supplemental Table 1, Supplemental Digital Content, http://links.lww.com/MSS/C983, Ingredients in the 500-kcal peanut and high carbohydrate snacks). At the start of each week, participants picked up a cooler containing all seven snacks for the week and returned the cooler from the previous week. Participants were asked to eat all their snacks but to return anything not consumed, so it could be weighed and recorded. The timing of the consumption of the snack was not controlled, but participants recorded the time of consumption on a daily snack tracking log and noted if they were consumed all at once or at two different times. Logs were returned to the researchers every week during RT sessions.

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The participants were blinded to the macronutrient content of their snacks and informed that both snack groups had the same potential to promote healthy weight gain. This was done to minimize a possible effect on participant compliance and/or of adjusting their regular diet by recognizing or believing to recognize one of the snack groups as being better for weight gain. Additionally, the study's resistance trainers were blinded to group allocation to eliminate potential bias.

Participants were instructed to maintain their normal daily dietary intake. At BSL and weeks 3, 7, and 10, participants were asked to record all foods and beverages they consumed for three days (two weekdays and one weekend day); they were instructed to include their 500 kcal snack on their food intake record during the intervention. These records were analyzed using Food Processor Nutrition Analysis Software (2021, Version 11.9.14, ESHA Research, Salem, OR).

***Nutrition Counseling.*** On the first day of the intervention, all participants received a nutrition education handout, which provided weight gain tips (Sports, Cardiovascular, and Wellness Nutrition (SCAN) Weight Gain in Sport, 2016 (27)), and were provided standardized education by a RD (AS) on ways to enhance healthy weight gain. This included consuming 500 additional kcals per day, choosing nutrient-dense, high-quality foods and beverages, eating breakfast, eating every 2-3 hours throughout the day, and aiming for 15-25 grams of protein post-exercise. Additionally, at the 3-wk and 7-wk lab visits, if a participant had not demonstrated weight gain and/or lean mass gain, participants were interviewed about their diet and the additional 500 kcal snack, and healthy weight gain recommendations were reinforced.

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***Resistance Training.*** Participants completed a 10-wk, hypertrophy-focused RT protocol under the close supervision of an American College of Sports Medicine (ACSM) Certified Personal Trainer (JR) in the exercise training laboratory. Each training week consisted of three days of training. Each training day worked both the upper and lower body. Day 1 was a pressing-focused day in which upper body movements, such as the bench press, and lower body movements, such as squats, were prioritized. Day 2 was a pulling-focused day in which upper body movements, such as rows, and lower body movements, such as Romanian deadlift, were prioritized. Day 3 consisted of both pressing and pulling movements. A detailed description of training days can be found in Table 1. This training protocol was structured such that Days 1 and 2 could be done consecutively if needed. At least one rest day took place before the completion of Day 3. The organization of training and rest days was based on the participant's and the trainers' schedules. Training sessions of different participants were allowed to overlap as long as close supervision from the trainer could be maintained. Participants who were engaged in a sport or military training activity were instructed to maintain their normal training throughout the study. However, all participants were asked to not participate in RT outside of our intervention.

Participants documented their training volume in logs, annotating lifts performed, amount of weight lifted, and number of reps and sets. The following resistance exercises were performed with weight machines (Nautilus) and free weights: leg press, squat, leg extension, leg curl, calf raise, lat pulldown, seated row, Romanian deadlift, shoulder press, bench press, biceps curl, triceps extension, shoulder lateral raise, banded face-pulls, and sit-ups. Participants performed 8 to 10 sets of exercises per muscle group per week in accordance with a recently published position stand of the International Universities Strength and Conditioning Association (IUSCA)

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(28). The principle of progressive overload was used in conjunction with Repetitions in Reserve (RIR), an autoregulation strategy that allowed participants and trainers to modulate the loading of an exercise based on performance and fatigue rather than a set percentage or weight (29). At the end of each set, participants were asked how many more reps were needed to bring them to failure. For example, if a participant estimated that he/she could have completed three more reps before failure, then the designated intensity was 3 RIR. Evidence suggests that RIR is more accurate as proximity to failure decreases (or the intensity increases) and that lifts performed at 3 RIR (or less), for ≤12 reps, have the greatest precision (29, 30). Prior to each lifting session, participants performed a warmup consisting of 3 to 5 mobility exercises followed by 2 to 3 warmup sets on each exercise; this excluded movements in which the muscle group would have been adequately warmed-up throughout the workout, i.e., participants did not warm-up for triceps extensions after doing bench press and shoulder press.

The RT protocol began on week 1 working the major muscle groups 5 to 6 times per week at 3 to 5 RIR. Workout volume increased until week 4 in which participants then executed 8 to 10 sets per week per muscle group at 3 RIR. On weeks 7 and 8, participants trained at 1 to 2 RIR. Week 9 was a deload week in which participants engaged in 5 to 6 sets per week at 3 to 5 RIR. During week 10, participants engaged in 8 to 10 sets per week per muscle group at 0 RIR (excluding Romanian deadlift). Participants were encouraged to adhere to prescribed reps. However, the concept of RIR stops was utilized to account for the possible over/underestimation of loading that occurs when utilizing RIR (31).

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The rest period between sets lasted 1 to 2 minutes. Participants were considered compliant if they finished 85% of the 30 workouts over the 10 weeks. Upper and lower body strength was assessed using a 1-rep max barbell back squat and barbell bench press performed at weeks 2 and 10 of the training protocol.

**Experimental Assessments**

All measurements were performed in the morning between 6am and 9am after an 8-hour fast (no caffeine) with the participants instructed not to engage in heavy exercise for 36 hours prior to testing and to adequately hydrate the evening before (32). Participants were provided with two 500 ml water bottles and were asked to consume one water bottle the night before and one water bottle the morning of their appointment.

**Dual-Energy X-ray Absorptiometry (DXA) and Urine Specific Gravity (USG)**

DXA was used to assess total body composition at four time points during the study (BSL, weeks 3, 7, 11) and was performed on a Lunar iDXA (GE Healthcare, Madison, WI, USA). Stature was obtained with the head in the Frankfort plane using a stadiometer (Welch Allyn Scale-Tronix 5002, Milwaukee, WI), and weight measured (nearest 0.1 kg) with participants wearing lab-provided lightweight gym T-shirt and shorts. Quality assurance procedures were performed before each DXA. DXAs were primarily performed by one certified radiology technician. The coefficient of variation for the DXA total body scan for repeated measurements was 0.025% for FM, 0.625% for fat-free mass, and 0.572% for TBM.

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Participants were required to empty their bladder and also provide a urine sample immediately before the DXA scan. The urine was evaluated for hydration status via specific density using a refractometer (Fisherbrand Analog Clinical). A pregnancy test was performed on the urine sample for female participants.

**Circumferences**

Circumferences were obtained in duplicate according to the procedures outlined by the International Standards for Anthropometric Assessment (33) at the following anatomical sites on the right side of the body by primarily one researcher (AS; pre and post measurements were always done by the same researcher): calf, hip, waist, mid-upper arm relaxed, and biceps relaxed and flexed. A third measure was taken if the duplicates varied by >0.5 cm.

**Blood Collection and Analyses**

A blood sample was obtained via venipuncture at four timepoints during the study (BSL, weeks 3, 7, 11) for a complete blood count (CBC), lipid panel, and analysis of TSH and fasting glucose concentrations by a commercial laboratory (LabCorp, Roanoke, VA). Samples sat at room temperature for 15-25 minutes before centrifugation at 4°C for 13 minutes at 3500 rpm.

**Resting Blood Pressure and Heart Rate**

BP and HR were measured on the right arm using automatic BP monitors (Omron Intellisense, Kyoto, Japan; GE Dinamap V100, Milwaukee, Wil) after a 10-minute rest period with the subject in a seated position. The BP measurements were repeated every three minutes until within-session stability was achieved (≤6 mm Hg on three sequential measurements up to six measures).

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## Aerobic Fitness

Using a motor-driven treadmill (Woodway, Waukesha, WI, USA) at pre- and post-intervention, a graded exercise test was performed to assess aerobic fitness via indirect calorimetry (800 lpm pneumotach, Parvo Medics TrueOne 2400, Sandy, UT, USA). Standard calibration procedures were conducted beforehand. HR was measured continuously using a portable HR monitor (Polar, Bethpage, NY, USA). The test began with a walking economy test in which the participant walked during three 4-minute stages at 2.5, 3 and 3.5 miles per hour (mph). The grade was increased by 2.5% if the participant's HR was greater than 120 beats per minute (bpm) or increased by 5% if HR was less than 120 bpm, which the participant sustained for one minute. Next, the workload was increased each minute by increasing treadmill speed or grade by 0.5 mph or 2.5%, respectively, until the participant reached volitional exhaustion. Rating of Perceived Exertion (RPE) was assessed during the last 15 seconds of each stage using the Borg Scale (6-20) (34). The highest 15-second $VO_2$ and respiratory exchange ratio (RER) achieved in the final two minutes of exercise were recorded as the maximum values. To qualify as a valid maximum test, participants had to meet two of the four following criteria: (1) maximum HR within 10% of predicted maximum [208 – (0.7 x age)] (35); (2) RPE ≥18; (3) RER ≥ 1.10; (4) evidence of a $VO_2$ plateau which was determined using the last 45 seconds of each stage and a ≤1.5 $mL \cdot min^{-1} \cdot kg^{-1}$ change in $VO_2$.

## Physical Activity

Physical activity was monitored using a smart watch (Garmin Venu Sq, Olathe, KS, USA), which recorded HR, step counts, and estimated energy expended during structured exercise or physical activity; this was used to help estimate the participants' total energy

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expenditure. Participants wore the smart watch for three days at BSL and at weeks 3, 7, and 10 of the intervention in conjunction with their 3-day food intake records. They also completed a separate 3-day exercise log during these timepoints.

**Post-Intervention Questionnaires**

Post-intervention, participants completed a snack questionnaire to assess snack acceptability and an open-ended exit survey to assess diet, exercise, and sleep habits during the 10-wk intervention compared to BSL (Supplemental Table 2, Supplemental Digital Content, http://links.lww.com/MSS/C983, Post-intervention questionnaire results).

**Statistical Analyses**

Descriptive statistics were generated for BSL characteristics and exit survey responses in the overall sample and by snack group (PNT vs CHO), including means and standard deviations (SD) for normally distributed continuous variables, medians and minimums and maximums for non-normally distributed continuous variables, and counts and percentages for categorical and ordinal variables. Distributions of all variables were compared across snack group, with additional comparisons across sex. Group comparisons of continuous data relied on two sample t-tests or Wilcoxon rank-sum tests, as appropriate. Comparisons of categorical variables between snack groups and sex were evaluated using Chi-square tests or Fisher's exact tests, as appropriate. Additionally, descriptive statistics for all outcomes of interest were performed at BSL and weeks 3, 7, and 11. Visualizations of estimated changes over time were performed in the various outcomes by snack group.

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Linear mixed effects models were performed for each outcome of interest collected at all four timepoints (BSL, weeks 3, 7, and 11). Models regressed each outcome on snack group (regardless of time), time (regardless of snack group), and the two-way interaction comparing changes over time by snack group (snack group * time). Similarly, additional linear mixed effects models were performed regressing each outcome of interest on sex, time, and the two-way interaction to examine changes over time by sex (sex * time). General linear models were performed to examine the association between snack group and change scores from BSL to post-intervention in outcomes collected at only these two timepoints. For these models, change scores for each of the outcomes were regressed solely on snack group. Estimates, standard errors, 95% confidence intervals, and the corresponding *P*-values were generated and reported. Paired t-tests were performed to examine pre to post changes in the overall sample for circumferences and $VO_2$max tests, as well as nutrient differences regarding participants who achieved or did not achieve the weight gain goal. Missing data for all variables of interest was minimal (<5%). Thus, analyses are based on available and observed data. Data are presented as mean ± SD unless stated otherwise. Statistical significance was taken at the 0.05 level and did not adjust for multiplicity. All descriptive statistics and mixed models were performed using R Version 4.1.0 (2021-05-18) and SAS V9.4 (SAS Institute Inc., Cary, NC).

## RESULTS

Thirty-five participants were enrolled in this study from November 2021 to November 2022. The final analytical sample consisted of 32 participants who completed the intervention (91% retention). Figure 2 summarizes recruitment and enrollment information. Of the 32, 19

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men (24.6±5.5 y, 75.4±8.7 kg, 178±8 cm, 23.5±2.6 kg/m$^2$) and 13 women (24.8±5.8 y, 60.4±9.3 kg, 166±6 cm, 21.9±3.3 kg/m$^2$) participated. The following sport and military athletes were represented: college division II soccer (N=1), club [lacrosse (N=1), fencing (N=1), triathlon (N=1), volleyball (N=1)], recreational [distance runner (N=2), mountain biker (N=1), rock climber (N=1), martial arts (N=3), functional fitness (N=13)], active-duty Navy (N=4), Army National Guard (N =1), Marine Corps Reserve (N=1), and first responder (N=1).

Half of the participants (N=16 each) were randomized to the PNT and CHO groups. The BSL demographic and physical characteristics were not different between groups and are summarized in Table 2. Sample sizes of 16 participants per group (overall 32 participants) were determined to yield more than 90% power to detect a 2.27 kg weight gain over 10 weeks (large effect), assuming a SD of 1.1 kg, based on a paired t-test. Specific variables were selected for sex comparison, based on men's typical larger frame, higher LBM, and higher energy needs compared to women.

Additionally, of the 32 participants, 15 (10 men, 5 women) were categorized into the high RT experience group, and 17 (9 men, 8 women) into the low RT experience group. After randomization, in the PNT group, five (4 men, 1 woman) had high RT experience, and in the CHO group, 10 (6 men, 4 women) had high RT experience. The participants' RT history averaged 3 to 5 years. The Baecke questionnaire indicated their main sport was between medium and high intensity that they engaged in about 4 hr/wk for 8 months/yr.

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## Weight Gain Intervention

***Dietary Intake.*** Snack compliance was 98.22±0.02% and 96.34±0.04% for the PNT and CHO groups, respectively. Daily energy and macronutrient intakes from the 3-day food intake records at BSL and weeks 3, 7 and 10 by snack group and sex are reported in Table 3. CHO and fat intake averaged 45-46% and 36-39% of total kcal/day, respectively (N=31). There were no differences over time by group ($P$=0.090) for protein intake, which averaged between 1.5 and 1.8 g/kg/day (15-16% of total kcal/day). Men, however, had higher overall energy, protein, and fat intakes than women (Supplemental Table 3; $P$=0.005, $P$=0.005, $P$=0.003, respectively, Supplemental Digital Content, http://links.lww.com/MSS/C983, Linear mixed effects model results for examining change in body composition, dietary intake, and physical activity over time by sex of participants).

***Nutrition Counseling***. Due to suboptimal weight gain (<0.23 kg/week), nine participants (6 PNT, 3 CHO) were questioned about their eating habits and snack consumption and received further nutrition education targeting healthy weight gain at their 3-wk lab appointment and three participants (2 PNT, 1 CHO) at their 7-wk lab assessment.

***Resistance Training, Aerobic Fitness, and Physical Activity.*** Participants (N=31) completed 28±2 total workouts over the 10-wk intervention. The 1-rep max barbell bench press and back squat increased from week 2 to 10 by 5.7±3.9 kg ($P$<0.001) and 9.6±4.9 kg ($P$<0.001), respectively, with no difference by group (Table 3). The participant who finished the intervention through week 7 completed 21 total workouts and did not perform strength assessments during week 10. From paired t-tests, relative ($mL \cdot min^{-1} \cdot kg^{-1}$) $VO_2$max values

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decreased ($P$=0.015) due to the increase in TBM, yet absolute ($L \cdot kg^{-1}$) $VO_2$max values remained unchanged ($P$=0.195). As summarized in Table 3, there were no differences by snack group over time for participants' daily steps ($P$=0.188).

**Experimental Assessments**

***Body Weight and Body Composition Changes.*** TBM increased 2.2±1.3 kg (range of -0.9 to 4.4 kg) with 1.5±1.1 kg (range of -0.5 to 3.5 kg) as LBM after 10 weeks (N=32) (Figures 3a and b, Table 4). For each week throughout the intervention, TBM, LBM, FM, and body fat percent increased by an average of 0.24 kg, 0.16 kg, 0.08 kg, and 0.04%, respectively ($P$<0.001, $P$<0.001, $P$<0.001, $P$=0.169, respectively). The PNT group gained less TBM than the CHO group (1.6±1.1 kg vs 2.7±1.2 kg, respectively, $P$=0.007) but gains in LBM were not different between groups (1.2±1.1 kg vs 1.9±1.0 kg, $P$=0.136) (Figure 4). FM increased more rapidly in the CHO group during the 10 weeks compared to the PNT group ($P$= 0.012). There were no differences between groups or in the group by time interaction for body fat percent over 10 weeks ($P$=0.277, $P$=0.130, respectively). Paired t-tests demonstrated significant changes from BSL to post-intervention in waist, hip, calf, and arm circumferences ($P$<0.001, $P$=0.002, $P$=0.035, $P$<0.001, respectively), but these changes were not different between the groups (Supplemental Table 4, Supplemental Digital Content, http://links.lww.com/MSS/C983, BSL to post-intervention results for circumferences and health outcomes of participants assigned to the PNT and CHO snack groups). Men and women had a TBM gain of 2.0±1.3 kg and 2.4±1.1 kg ($P$=0.310) with a LBM gain of 1.3±1.1 kg and 1.9±1.0 kg ($P$=0.294), respectively, over 10 weeks. USG averaged 1.014±0.007, 1.008±0.006, 1.016±0.010, and 1.012±0.008 at BSL, weeks

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3 and 7, and post-intervention, respectively, with the USG for most participants falling within the normal range for euhydration of <1.020 (2).

As shown in Figure 3a, five participants gained ≥2.27 kg (~0.23 kg/wk) by week 3. Only one participant lost these gains by week 7 and did not regain them by the end of the study. Eleven more participants gained ≥2.27 kg by week 7, for a total of 15 participants by week 7. Of these 15 participants, two individuals lost these gains by the end of the study, and it took two participants until the end of the study to gain ≥2.27 kg.

From paired t-tests (data not shown), the 15 participants who gained ≥2.27 kg by week 7 reported consuming 275±568 kcal/day more than BSL with no apparent change in grams of protein/kg or grams of CHO/kg (0.02±0.5, $P$=0.272, and 0.2±1.2, $P$=0.135, respectively) compared to those who gained less than predicted ($P$=0.038) and reported consuming less energy, protein, and CHO compared to BSL (-163±547 kcal, -0.2±0.4 g/kg, -0.3±0.9 g/kg, respectively). At the end of the study, the 15 participants who gained ≥2.27 kg reported consuming 602±637 kcal/day and 0.9±0.9 grams of CHO/kg/day more than BSL ($P$=0.011 and $P$=0.010) with relatively no change in grams of protein/kg/day (0.1±0.5 g/kg, $P$=0.948) compared to those who gained less than predicted and reported minimal increases in energy, protein, and CHO compared to BSL (43±518 kcal, 0.1±0.5 g/kg, 0.02±0.8 g/kg, respectively).

***Health Outcomes.*** *S*erum lipids, fasting blood glucose, resting BP, and HR were not affected by the weight gain intervention (Supplemental Table 4, Supplemental Digital Content, http://links.lww.com/MSS/C983 ).

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## DISCUSSION

This study evaluated the efficacy of current sports nutrition recommendation of a modest energy surplus from either peanut (PNT) or high-CHO snack foods, in combination with rigorous RT, over 10 weeks for promoting healthy weight gain in athletic men and women. After the study intervention period, we observed that TBM increased 2.2±1.3 kg with 68% as LBM. Providing excess energy from peanut foods was less efficient for promoting gains in TBM.

Our study's average weekly TBM gains of 0.22 kg were less compared to the majority of previous studies which demonstrated gains of 0.29-0.6 kg/wk with a daily energy surplus ranging from 500 to above 2000 kcal (14-18, 20, 21). However, they are similar to those of Kreider et al. (0.20 kg/wk) (15) and Spillane and Willoughby (0.18 kg/wk) (18), who provided daily energy surpluses of 760 kcal (15) or 1,101 kcal (18) as CHO powders mixed with fluid. The LBM gains of 1.5±1.1 kg in our study, however, were greater than several studies including those of Taguchi et al. (1.15-1.26 kg) (20), the aforementioned CHO group of Spillane and Willoughby (0.25 kg) (18), and the maltodextrin and Gainers Fuel 1000 supplement groups of Kreider et al. (0.67 kg) (15) despite the provision of a greater energy surplus ranging from 760-1,400 kcal/day and a protein intake range of 1.0-2.7 g/kg. Our LBM gains were similar to those of Garthe et al. who also provided ~500 kcal surplus of energy in addition to nutrition counseling (1.6 kg) (16), but slightly less than Spillane and Willoughby (2.3 kg) (18), Bartels et al. (2.4-3.5 kg) (14), Kreider et al. (2.02 kg in the Phosphagain group) (15), Rodriguez et al. (2.4 kg) (21), and Rozenek et al. (2.9-3.4 kg) (17). Higher LBM gains in Rozenek et al.'s

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participants compared to our study may be explained by the fact that they were novice resistance trainers, who may experience greater gains when beginning an RT regimen (8).

Most of the aforementioned studies that reported higher TBM gains included a daily energy surplus that ranged from ~650 – 2,020 kcal/day, which was considerably more than the 500 kcal/day provided in the current study. Exceptions include the studies of Garthe et al. (16), Bartels et al. (14), and Rodriguez et al. (21) that initially assigned a ~500 kcal/day surplus but adjusted energy intake upward during the intervention for participants who were not achieving the desired weight gain. We did not adjust individual excess energy levels, because we wanted to evaluate the individual response to the recommended standard 500 kcal/day surplus. We believe the greater energy surpluses may be one reason for the differences in TBM and LBM gain in previous studies compared to our study. Additionally, Garthe et al. (16) demonstrated the importance of weekly nutritional counseling for enhancing weight gain in athletes. Our study provided nutrition counseling initially but only provided additional counseling at the 3-wk and 7-wk assessments for participants who did not attain expected weight gain. Though counseling was standardized, it was not specifically individualized to the degree that would be done by a sports RD, as done in the study by Garthe et al (16).

We used 3-day self-reported food intake records to evaluate if our participants were fully compliant with maintaining their typical diet along with study-provided 500 kcal/day snack. Error in self-reported food intake records is well-recognized, and overall, individuals tend to underreport by 15% (36). Therefore, our participants may have underreported intake at some or

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all timepoints, as their total energy intake did not reflect a 500 kcal surplus above that of their BSL caloric intake. Even so, the CHO group significantly increased their energy intake during the intervention at weeks 3 and 7 compared to the PNT group.

Though increasing total dietary energy intake may be the most important nutritional factor for promoting healthy weight gain (8), there is little research supporting which macronutrient or combination of macronutrients should comprise these additional energy needs. Our participants reported consuming between ~2400 and 3300 kcal/day at all timepoints, which should have provided sufficient additional energy for training, optimization of MPS, and other tissue repair (13). Based on anecdotal evidence, Kleiner (8) reported that total dietary energy, specifically CHO, is the most-important nutritional factor affecting LBM gain. Our results confirmed this recommendation as noted by the significant increase in g/kg/day of both energy and CHO at weeks 3 and 7 and the slightly but not significantly greater LBM gain in the CHO compared to the PNT group. Conversely, many athletes presume the energy surplus should be from protein alone, which is not supported in the literature (17, 18). Two recent reviews (37, 38) recommend daily protein intakes of ≥1.6 g/kg/day to contribute to LBM gains from RT, which in our study was consumed by the CHO group at all timepoints, compared to the PNT group, who averaged slightly less (1.5 g protein/kg/day at BSL, weeks 3 and 7). These differences, however, were not statistically significant. Additionally, it is important to note that participants assigned to both snack groups met current sports nutrition guidelines of 1.2 to 2.0 g protein/kg/day, which may explain why there was no difference in LBM gains between the groups.

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Although we originally hypothesized there would not be a difference between groups for TBM gain when excess energy was provided by mostly protein and healthy fats from peanuts vs simple CHO obtained from whole foods, our data demonstrate that TBM gain was significantly less than predicted in the PNT compared to the CHO group. This finding may be due to macronutrient differences between our snack groups or a specific quality of peanuts and possibly other tree nuts. Previous studies in non-athletes have also reported that consumption of peanuts or nuts does not result in weight gain (39-43). The protein content of peanuts may increase satiety, which may have influenced the PNT group's ability to consistently overfeed without compensation (44). In addition, it has been demonstrated that the energy content of peanuts may not be completely digestible (45, 46) and therefore not fully bioavailable. Additionally, there was no difference in snack acceptability between the groups. Thus, a hyperenergetic peanut-based regimen might be most appropriate for athletes who want to gain LBM but not necessarily FM.

For optimal healthy weight gain, a strenuous RT program is known to be necessary to stimulate lean tissue accretion (6). Similar to our study, previously mentioned studies also employed full body exercises three to four times per week (16-18, 21), which were sufficient to stimulate muscle hypertrophy (13). Our RT protocol, designed to maximize hypertrophy, was rigorous and stimulated significant gains in TBM, LBM, as well as increased strength, demonstrated by a 13.7±9.7% and 11.3±8.4% increase in the total weight lifted for the barbell back squat and bench press 1-rep max, respectively. It is possible, however, that the program may have been either too rigorous

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or not rigorous enough for some of our participants. Brook et al. (47) reported that muscle gains are seen in the first 4 to 6 weeks of RT but then taper, which we observed in the 7-wk TBM and LBM plateau of the PNT group.

To supply the energy excess, most research studies that examined hyperenergetic feeding in combination with RT used UPF (14, 15, 17, 18, 21), which in non-athletes are associated with an increased risk of metabolic syndrome and all-cause mortality (23). The strengths of our study include use of minimally processed whole foods, which are nutritionally superior to UPF due to containing less added sugar, sodium, and fat, and no preservatives and other additives, and providing more fiber, vitamins, minerals, and phytochemicals. Additionally, our results demonstrate healthy weight gain, as blood lipids, fasting blood glucose, and BP did not increase throughout the intervention despite weight gain. Conversely, Gentile et al. (48), who used UPF to overfeed participants, observed an elevated BP after intentional weight gain. Unfortunately, many previous studies neglected to evaluate BP, which may not be expected to increase in athletes, since they are more physically active, but it is still important to assess. These results can encourage sports professionals to recommend the whole food approach to their athletes to support healthy weight gain.

Additional strengths include robust screening for enrollment, where all participants were screened for energy and total daily protein intake. We had proper blinding controls, dietary records with concurrent accelerometry before and at three timepoints during the intervention, and appropriate standards of practice for DXA outcomes (i.e., euhydration status, no prior RT for 36

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hours, and one primary DXA technician). Further strengths include a high retention rate, supervised RT program, and high snack adherence. Lastly and maybe most importantly, almost half of our participants were female athletes, and the data provide evidence these women were able to gain weight and LBM as effectively as the men in the study.

Our study has several limitations, including the randomization of participants which resulted in an uneven distribution of high vs low RT experience between the groups. However, this does not appear to have affected our results, as there was no difference in LBM gains between groups, although individuals with more RT experience are likely to gain less LBM (8). A second limitation is that we provided participants with only a 500 kcal daily snack and not all their daily provisions. However, the intent of this study was to evaluate current sports nutrition recommendations for healthy weight gain in a real-world context and not to tightly control all feeding aspects. A final limitation is that we, like others, did not include a control group, who could either participate in the RT program but not receive daily snacks or receive snacks for excess energy but not perform RT. These control groups might have assisted our (or previous studies) evaluation of the contribution of either diet or RT to healthy weight gain. Though the heterogeneity of our athletic population could be considered a limitation, we believe this sample population is representative for real life practice by sports dietitians, who provide performance nutrition counseling to athletes involved in a variety of sports. Future randomized controlled trials should examine the efficacy of weight gain with a controlled energy surplus to better understand the considerable inter-individual variability and determine the energy cost and macronutrient composition for muscle tissue accretion in the athletic population.

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## CONCLUSIONS

The study results suggest that the addition of 500 kcal/day from whole foods/snacks, in combination with a rigorous RT program, promotes weight gain of ~0.22 kg/wk, primarily as LBM, over 10 weeks in both male and female athletes. However, snack macronutrient content may impact the effectiveness of this regimen.

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## Conflict of Interest and Source of Funding

The views and information presented are those of the authors and do not represent the official position of the U.S. Army Medical Center of Excellence, the U.S. Army Training and Doctrine Command, or the Department of the Army, Department of Defense, or U.S. Government. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. This study was funded by a grant from the Peanut Institute Foundation. A supplemental data set and code are available at: https://figshare.com/s/d595bf8a12ec9cd362c0.

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## FIGURE LEGENDS

**Figure 1.** Overview of the study design and procedures

Participants ate a 500 kcal snack every day (d) in combination with supervised resistance training 3 d per week (wk) for 10 weeks. Before baseline (BSL), diet and exercise habits were screened. At BSL, weeks 3 and 7, and post-intervention, body composition via dual-energy x-ray absorptiometry (DXA), and specific blood analyses were assessed. At BSL, weeks 3, 7, and 10 of the intervention, 3-d food intake records were collected and a smart watch for step count was worn. At BSL and post-intervention, blood pressure, heart rate, circumferences, and $VO_2$max were measured. At weeks 2 and 10, strength assessments of a 1-repetition maximum barbell bench press and back squat were conducted. Participants tracked their snack consumption and exercise every day. Questionnaires regarding snack acceptability and ease of use, as well as diet, exercise, and sleep habits throughout the intervention, were administered post-intervention.

**Figure 2.** Participant flow diagram

Enrollment, allocation, and analysis

**Figure 3a.** Individual body composition changes (D) in TBM from baseline (N=32)

Data reported as means. TBM: total body mass.

**Figure 3b.** Individual body composition changes (D) in LBM from baseline (N=32)

Data reported as means. LBM: lean body mass.

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**Figure 4.** Body composition changes from baseline by snack group (N=32)

*Significant difference for TBM (*P*=0.007) and FM (*P*=0.012) from mixed model results. Data are presented as median (range). PNT: peanut, CHO: carbohydrate, TBM: total body mass, LBM: lean body mass, FM: fat mass.

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**SUPPLEMENTAL DIGITAL CONTENT**

**SDC 1:** Supplemental Digital Content.docx

**SDC 2:** Supplemental data set (external link)

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Figure 1. Overview of the study design and procedures

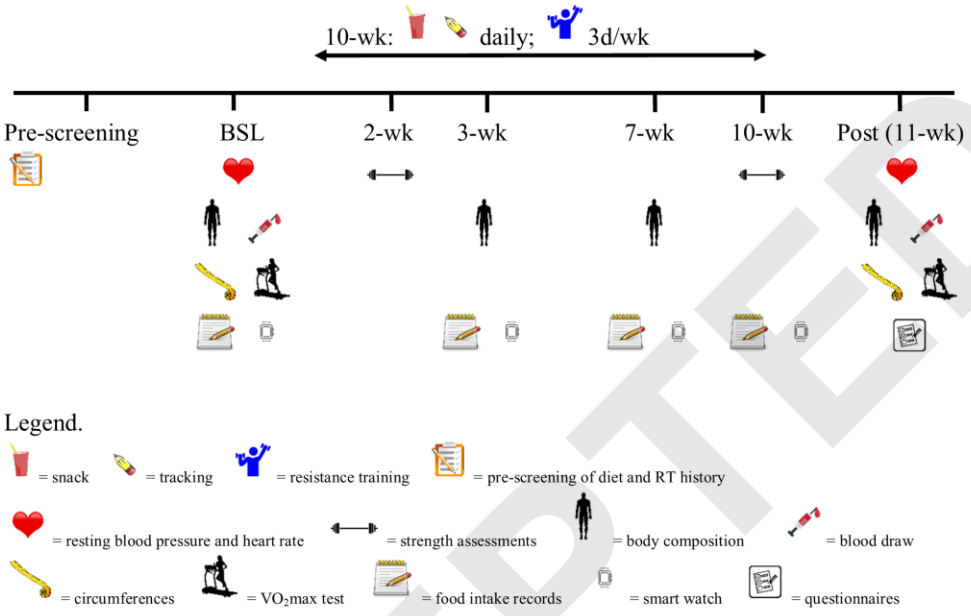


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Figure 2. Participant flow diagram

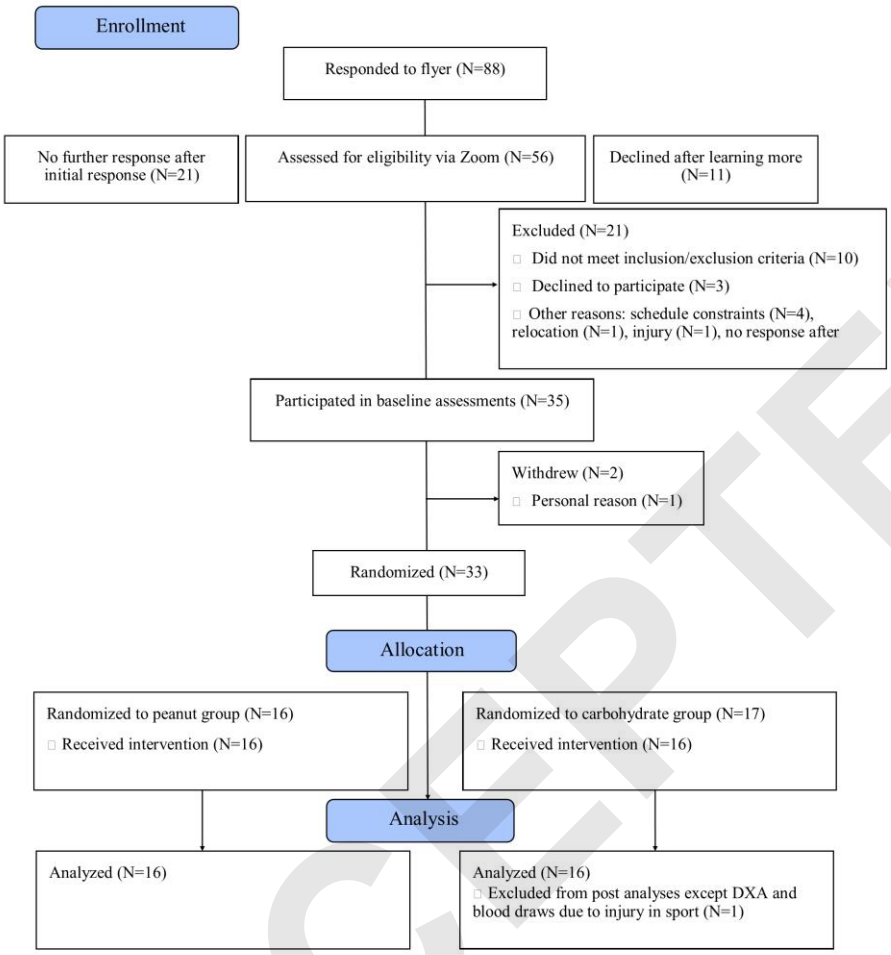


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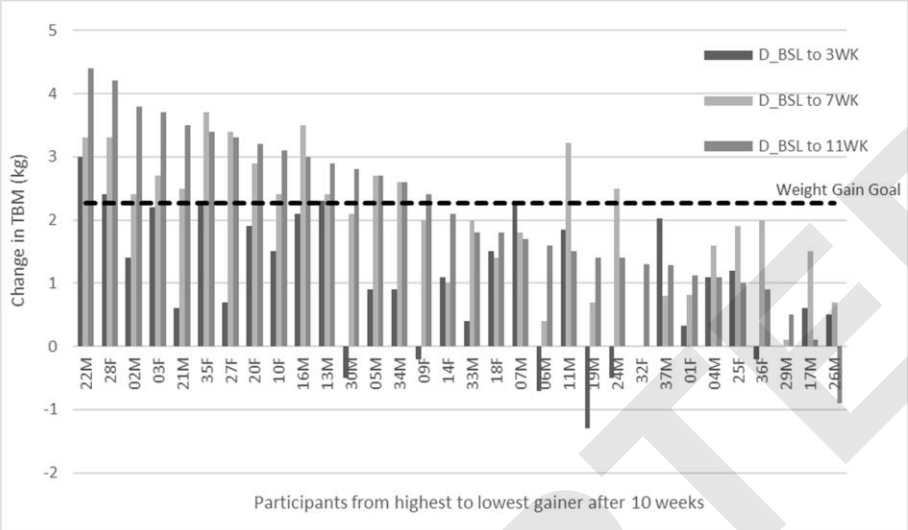
Figure 3a.

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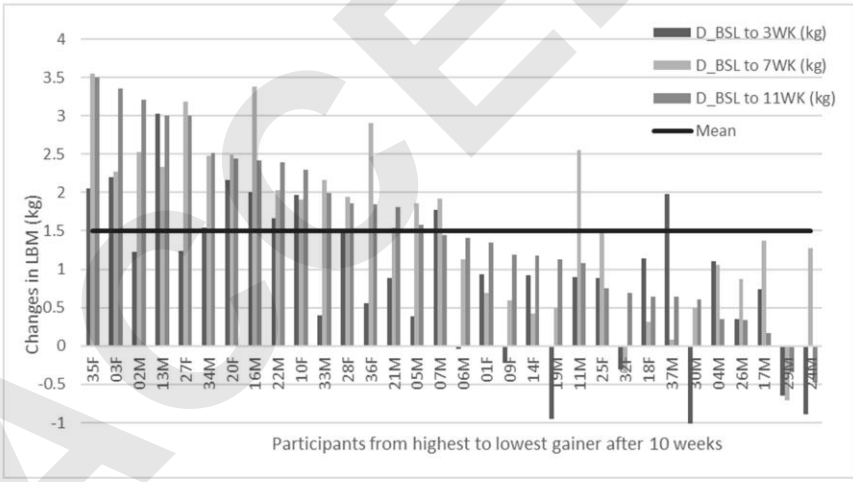


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Figure 4. Body composition changes from baseline by snack group (N=32)

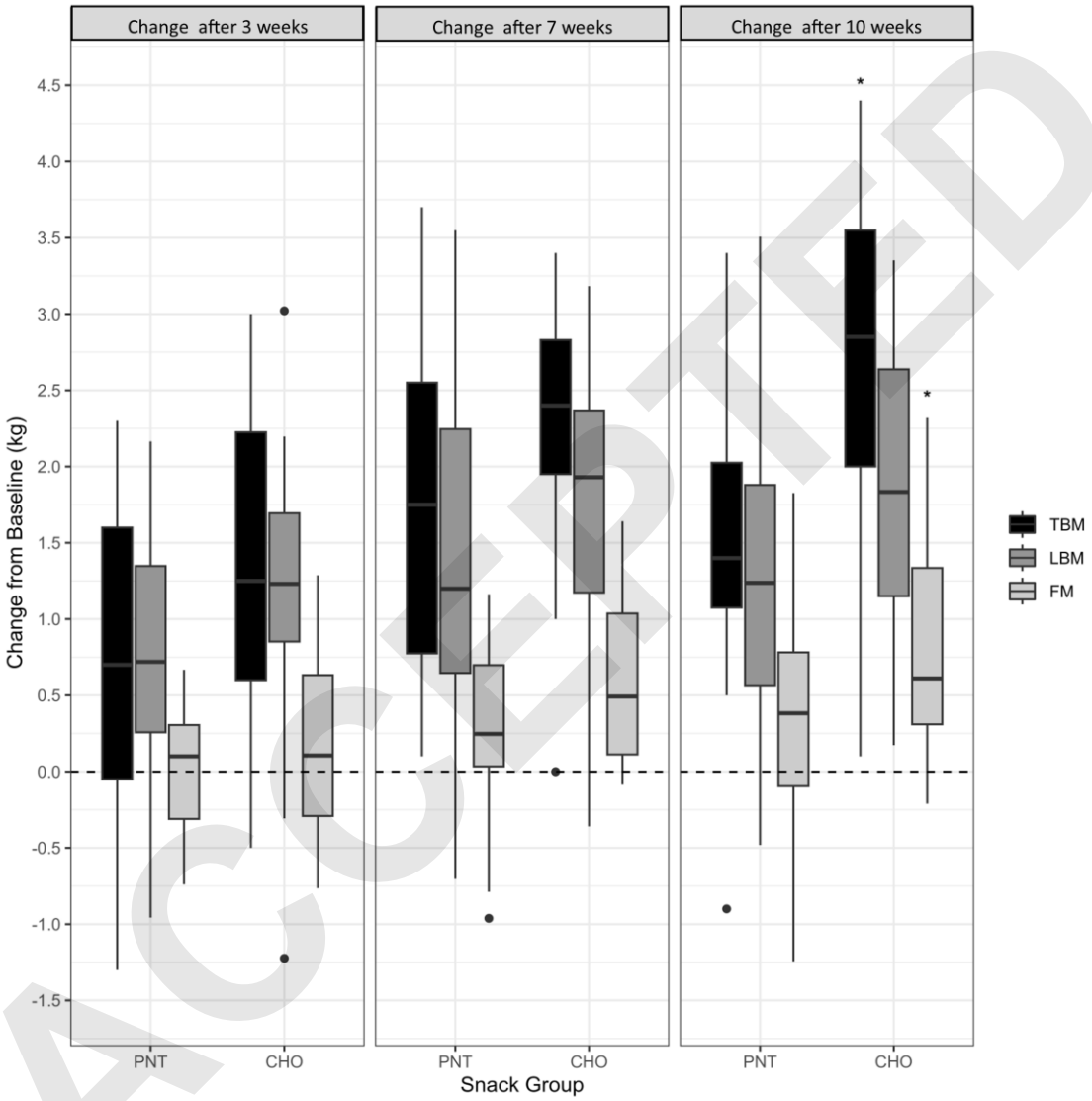


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**Table 1.** Resistance Training Protocol

| | Week | 1 | 2 | 3 | 4 | 5 & 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Exercises** | | | | | **(Sets) x (Reps) @ RIR** | | | | |
| **ay 1** | BB Squat | 2 x 12 @ 3-5 | 1 x 1 @ 0 | 3 x 12 @ 3 | 4 x 12 @ 3 | 4 x 10 @ 3 | 4 x 8 @ 2 | 4 x 8 @ 2 | 2 x 8 @ 3-5 | 1 x 1 @ 0 |
| | Leg Extension | 2 x 12 @ 3-5 | 2 x 12 @ 3 | 2 x 12 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 2 | 3 x 12 @ 2 | 2 x 12 @ 3-5 | 3 x reps @ 0 |
| | BB Calf Raise | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 2 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| | BB Bench Press | 2 x 12 @ 3-5 | 1 x 1 @ 0 | 3 x 12 @ 3 | 4 x 12 @ 3 | 4 x 10 @ 3 | 4 x 8 @ 2 | 4 x 8 @ 2 | 2 x 8 @ 3-5 | 1 x 1 @ 0 |
| | Push Up | 2 x reps @3-5 | 2 x reps @ 3 | 2 x reps @ 3 | 3 x reps @ 3 | 3 x reps @ 3 | 3 x reps @ 2 | 3 x reps @ 1 | 2 x reps @ 3-5 | 3 x reps @ 0 |
| | DB Shoulder Press | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 12 @ 3 | 4 x 12 @ 3 | 4 x 10 @ 3 | 4 x 10 @ 2 | 4 x 10 @ 2 | 3 x 10 @ 3-5 | 4 x reps @ 0 |
| | DB Standing Triceps Ext | 2 x 15 @ 3-5 | 2 x 15 @ 3 | 4 x 15 @ 3 | 3 x 15 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 2 | 3 x 12 @ 1 | 2 x 12 @ 3-5 | 3 x reps @ 0 |
| | Sit-up | 3 x reps @ 3-5 | 3 x reps @ 3 | 4 x reps @ 3 | 5 x reps @ 3 | 5 x reps @ 3 | 5 x reps @ 2 | 5 x reps @ 1 | 3 x reps @ 3-5 | 5 x reps @ 0 |
| **ay 2** | BB Romanian Deadlift | 2 x 12 @ 3-5 | 2 x 12 @ 3 | 3 x 12 @ 3 | 4 x 12 @ 3 | 4 x 10 @ 3 | 4 x 10 @ 2 | 4 x 10 @ 2 | 2 x 10 @ 3-5 | 4 x 10 @ 2 |
| | Leg Curl | 2 x 12 @ 3-5 | 2 x 12 @ 3 | 2 x 12 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 2 | 3 x 12 @ 2 | 2 x 12 @ 3-5 | 3 x reps @ 0 |
| | Lat Pull-Down | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 12 @ 3 | 5 x 12 @ 3 | 5 x 10 @ 3 | 5 x 10 @ 2 | 5 x 10 @ 2 | 3 x 10 @ 3-5 | 5 x reps @ 0 |
| | Machine Row | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 12 @ 3 | 5 x 12 @ 3 | 5 x 10 @ 3 | 5 x 10 @ 2 | 5 x 10 @ 2 | 3 x 10 @ 3-5 | 5 x reps @ 0 |
| | DB Bicep Curl | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 10 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| | DB Shoulder Lateral Raise* | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| | Banded Face-Pull* | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| **ay 3** | Leg Press | 2 x 12 @ 3-5 | 2 x 12 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 3 | 3 x 10 @ 3 | 3 x 10 @ 2 | 3 x 10 @ 2 | 2 x 10 3-5 | 3 x reps @ 0 |
| | Romanian Deadlift | 2 x 12 | 2 x | 3 x | 3 x | 3 x | 3 x | 3 x | 2 x 8 | 3 x |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | @ 3-5 | 12 @ 3 | 12 @ 3 | 12 @ 3 | 10 @ 3 | 8 @ 2 | 8 @ 2 | @ 3-5 | 8 @ 0 |
| BB Calf Raise | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| Machine Chest Press | 2 x 12 @ 3-5 | 2 x 12 @ 3 | 3 x 12 @ 3 | 3 x 12 @ 3 | 3 x 10 @ 3 | 3 x 10 @ 2 | 3 x 10 @ 1 | 2 x 10 @ 3-5 | 3 x reps @ 0 |
| Machine Shoulder Press | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 12 @ 3 | 4 x 12 @ 3 | 4 x 10 @ 3 | 4 x 10 @ 2 | 4 x 10 @ 1 | 3 x 10 @ 3-5 | 4 x reps @ 0 |
| Lat Pull-Down | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 12 @ 3 | 5 x 12 @ 3 | 5 x 10 @ 3 | 5 x 10 @ 2 | 5 x 10 @ 1 | 3 x 10 @ 3-5 | 5 x reps @ 0 |
| Machine Row | 3 x 12 @ 3-5 | 3 x 12 @ 3 | 4 x 12 @ 3 | 5 x 12 @ 3 | 5 x 10 @ 3 | 5 x 10 @ 2 | 5 x 10 @ 1 | 3 x 10 @ 3-5 | 5 x reps @ 0 |
| DB Shoulder Lateral Raise* | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| Face Pull* | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| DB Bicep Curl† | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| DB Standing Triceps Ext† | 3 x 15 @ 3-5 | 3 x 15 @ 3 | 4 x 15 @ 3 | 5 x 15 @ 3 | 5 x 12 @ 3 | 5 x 12 @ 2 | 5 x 12 @ 1 | 3 x 12 @ 3-5 | 5 x reps @ 0 |
| Sit-Up | 3 x reps @ 3-5 | 3 x reps @ 3 | 4 x reps @ 3 | 5 x reps @ 3 | 5 x reps @ 3 | 5 x reps @ 2 | 5 x reps @ 1 | 3 x reps @ 3-5 | 5 x reps @ 0 |

RIR: repetitions in reserve; BB: barbell; DB; dumbbell; Ext: extension; reps: repetitions.

*Indicates a superset between shoulder lateral raise and banded face-pull

†Indicates a superset between DB bicep curl and standing triceps ext

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Table 2. Baseline characteristics of participants by snack group (N=32)

| | **PNT (N=16)** | **CHO (N=16)** | **Difference (*P*-value*)** |
|---|---|---|---|
| Age (y) ¥ | 23 (19, 39) | 22 (18, 28) | 0.119 |
| Sex (female), N (%) | 6 (38) | 7 (44) | 0.719 |
| Race, N (%) | | | 0.588 |
| Black/African American | 1 (6) | 0 (0) | |
| Asian | 1 (6) | 0 (0) | |
| White | 12 (75) | 12 (75) | |
| Hispanic/Latino | 2 (13) | 2 (13) | |
| Mixed† | 0 (0) | 2 (13) | |
| Total body mass (kg) | 67.7±12.7 | 70.4±10.1 | 0.522 |
| Height (cm) | 173.9±10.3 | 173.0±8.7 | 0.801 |
| BMI (kg/m$^2$) | 22.3±2.9 | 23.5±2.9 | 0.259 |
| Hemoglobin (g/dL) | 14.5±1.3 | 14.6±1.3 | 0.810 |
| TSH (uIU/ml) | 2.3±1.0 | 2.2±0.9 | 0.672 |
| Vegetarian, N (%) | 4 (25) | 0 (0) | 0.101 |
| Total Baecke score | 8.6±1.1 | 9.1±1.2 | 0.213 |
| work index | 2.2±0.6 | 2.4±0.8 | 0.344 |
| sport index | 3.6±0.6 | 3.6±0.6 | 0.652 |
| leisure index | 2.8±0.6 | 3.0±0.4 | 0.312 |
| $VO_2$max (mL/kg/min) | 48.8±6.7 | 52.2±7.4 | 0.184 |

Data reported as mean ± SD except for age, race, and vegetarian status. **P*-values were calculated based on two sample t-tests or Wilcoxon rank-sum tests for continuous variables and Chi-square tests or Fisher's exact tests for categorical variables. ¥Data reported as median (range). †Reported ≥2 race categories (percent total). PNT: peanut, CHO: carbohydrate, BMI: Body Mass Index, TSH: Thyroid Stimulating Hormone.

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Table 3. Reported dietary intake and physical activity with strength assessments for male and female participants assigned to the PNT (N=16; men: N=10, women: N=6) and CHO (N=15; men: N=8, women: N=7) snack groups

| | Group | BSL | 3-wk | 7-wk | 10-wk | *P*-value (snack group by time) | *P*-value (sex by time) |
|---|---|---|---|---|---|---|---|
| **Dietary intake** | | | | | | | |
| Total energy (kcal/d) | **PNT** | 2525±547 | 2395±515 | 2473±423 | 2826±707 | 0.590 | 0.812 |
| | Men | 2660±598 | 2310±609 | 2561±518 | 3064±778 | | |
| | Women | 2300±396 | 2522±340 | 2326±111 | 2431±332 | | |
| | **CHO** | 2986±917 | 3333±919* | 3113±572* | 3274±870 | | |
| | Men | 3620±640 | 3960±527 | 3506±387 | 3718±779 | | |
| | Women | 2261±581 | 2617±725 | 2664±382 | 2767±704 | | |
| CHO (g/kg/d) | **PNT** | 4.6±1.3 | 4.2±1.6 | 4.0±1.2 | 4.6±1.3 | 0.132 | 0.889 |
| | Men | 4.1±1.3 | 3.5±1.5 | 3.6±1.2 | 4.4±1.5 | | |
| | Women | 5.4±0.9 | 5.3±1.2 | 4.8±1.1 | 5.1±0.6 | | |
| | **CHO** | 4.7±1.2 | 5.3±1.4* | 5.1±0.9* | 5.5±1.3 | | |
| | Men | 5.4±1.2 | 5.9±0.9 | 5.3±0.8 | 5.9±1.4 | | |
| | Women | 3.9±0.5 | 4.7±1.6 | 4.9±1.0 | 4.9±1.1 | | |
| PRO (g/kg/d) | **PNT** | 1.5±0.5 | 1.5±0.5 | 1.5±0.3 | 1.8±0.5 | 0.090 | 0.504 |
| | Men | 1.5±0.4 | 1.4±0.5 | 1.4±0.4 | 1.8±0.6 | | |
| | Women | 1.7±0.6 | 1.8±0.5 | 1.5±0.3 | 1.7±0.3 | | |
| | **CHO** | 1.7±0.5 | 1.6±0.3 | 1.6±0.3 | 1.7±0.3 | | |
| | Men | 2.0±0.5 | 1.8±0.3 | 1.8±0.3 | 1.8±0.3 | | |
| | Women | 1.4±0.5 | 1.4±0.3 | 1.4±0.2 | 1.5±0.2 | | |
| Total fat (g/d) | **PNT** | 102±30 | 105±19 | 111±27 | 120±39 | **0.044** | 0.343 |
| | Men | 111±29 | 102±20 | 118±28 | 132±44 | | |
| | Women | 87±28 | 109±18 | 100±22 | 100±18 | | |
| | **CHO** | 125±5 | 149±50* | 124±30 | 127±32 | | |
| | Men | 152±41 | 179±47 | 140±29 | 135±29 | | |
| | Women | 95±27 | 115±29 | 105±21 | 117±35 | | |
| **Physical activity** | | | | | | | |
| Daily steps ¥ | **PNT** | 8,598 (2,299-18,217) | 7,664 (2,306-11,671) | 9,132 (3,365-13,058) | 7,926 (2,192-12,280) | 0.188 | 0.810 |
| | Men | 7,269 (2,299-18,217) | 7,346 (2,306-11,671) | 8,306 (3,365-13,058) | 7,979 (2,192-12,280) | | |
| | Women | 9,530 (6,290-15,622) | 8,708 (6,157-9,257) | 9,555 (4,019-13,000) | 7,503 (2,745-11,716) | | |
| | **CHO** | 8,999 (4,413-14,981) | 10,655 (4,928-16,551) | 9,059 (4,927-19,080) | 9,064 (4,395-21,255) | | |
| | Men | 8,154 (5,987-12,649) | 8,843 (4,928-16,551) | 9,181 (4,927-10,952) | 8,747 (4,395-19,131) | | |
| | Women | 12,267 (4,413-14,981) | 11,596 (5,449-16,208) | 8,279 (6,388-19,080) | 9,064 (7,989-21,255) | | |
| **Strength assessments†** | | | | | | | |
| 1-RM bench press (kg) | **PNT** | 60.1±25.3 | - | - | 65.5±24.9 | 0.715 | 0.784 |
| | Men | 75.2±17.3 | - | - | 80.7±16.4 | | |
| | Women | 34.9±12.5 | - | - | 40.2±11.8 | | |
| | **CHO** | 66.8±31.0 | - | - | 72.7±30.9 | | |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Men | 86.7±24.7 | - | - | 92.9±24.4 | | |
| | Women | 44.2±20.2 | - | - | 49.7±19.1 | | |
| 1-RM back squat (kg) | **PNT** | 82.7±28.6 | - | - | 93.3±27.9 | 0.209 | 0.488 |
| | Men | 98.9±18.7 | - | - | 109.1±18.7 | | |
| | Women | 55.7±20.6 | - | - | 67.1±19.6 | | |
| | **CHO** | 88.8±32.34 | - | - | 97.3±31.4 | | |
| | Men | 110.2±21.8 | - | - | 117.9±21.5 | | |
| | Women | 64.3±24.0 | - | - | 73.7±23.5 | | |

Data reported as mean±SD based on 3-day food and physical activity records with *P*-values calculated from linear mixed effects models. *Significant difference by snack group at same timepoint ($P<0.05$). ¥ Data reported as median (range). †General linear models were used to calculate *P*-values for strength assessments, which were performed at weeks 2 and 10 of the intervention. PNT: peanut, CHO: carbohydrate, BSL: baseline, d: day, PRO: protein, RM: repetition maximum.

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Table 4. Body composition of participants assigned to the PNT and CHO snack groups (N=32)

| | Group | BSL | 3-wk | 7-wk | Post | *P*-value (time) | *P*-value (snack group x time) |
|---|---|---|---|---|---|---|---|
| **Body composition** | | | | | | | |
| TBM (kg) | PNT | 67.7±12.7 | 68.5±12.3 | 69.5±12.5 | 69.3±12.5 | <0.001 | 0.007 |
| | CHO | 70.4±10.1 | 71.6±9.8 | 72.6±10.2 | 73.1±9.8 | | |
| | PNT+CHO | 69.0±11.4 | 70.0±11.1 | 71.1±11.3 | 71.2±11.3 | | |
| LBM (kg) | PNT | 49.9±11.0 | 50.7±10.6 | 51.4±11.0 | 51.2±10.8 | <0.001 | 0.136 |
| | CHO | 53.2±10.2 | 54.3±9.9 | 54.9±10.2 | 55.1±9.9 | | |
| | PNT+CHO | 51.6±10.6 | 52.5±10.3 | 53.2±10.6 | 53.1±10.4 | | |
| FM (kg) | PNT | 15.0±4.3 | 15.0±4.5 | 15.3±4.4 | 15.4±4.5 | <0.001 | 0.012 |
| | CHO | 14.2±4.4 | 14.4±4.5 | 14.8±4.6 | 15.1±4.6 | | |
| | PNT+CHO | 14.6±4.3 | 14.7±4.4 | 15.0±4.4 | 15.2±4.5 | | |
| Body fat (%) | PNT | 23.5±6.2 | 23.2±6.2 | 23.3±6.3 | 23.4±6.2 | 0.169 | 0.130 |
| | CHO | 21.4±7.1 | 21.2±7.0 | 21.6±7.0 | 21.7±7.0 | | |
| | PNT+CHO | 22.5±6.7 | 22.2±6.6 | 22.4±6.6 | 22.6±6.5 | | |

Data reported as mean±SD. *P*-values calculated from linear mixed effects models with significant difference ($P<0.05$). PNT: peanut, CHO: carbohydrate, BSL: baseline, TBM: total body mass, LBM: lean body mass, FM: fat mass.

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Supplemental Table 1. Ingredients in the 500-kcal peanut and high carbohydrate snacks

| PNT Snack | Ingredient | Amount | CHO Snack | Ingredient | Amount |
|---|---|---|---|---|---|
| PB Smoothie | PB<br>Banana<br>Whole milk | 4<br>9.5 g<br>8<br>0 g<br>2<br>27 g | Fruit Smoothie | Strawberries (frozen)<br>Blueberries (frozen)<br>Spinach (fresh)<br>Orange juice<br>Avocado<br>Olive oil<br>Sugar | 5<br>5 g<br>5<br>7.5 g<br>3<br>0 g<br>1<br>70 g<br>6<br>8 g<br>1<br>0.8 g<br>4<br>0 g |
| Trail Mix | Peanuts<br>PB Crunch cereal<br>Chocolate chips | 4<br>2 g<br>4<br>8 g<br>1<br>0 g | Trail Mix | Raisins<br>Corn Chex<br>Pretzels<br>Chocolate chips<br>Banana chips<br>Cranberries (dried) | 1<br>0 g<br>1<br>5 g<br>2<br>8 g<br>2<br>0 g<br>1<br>8 g<br>3<br>5 g |
| PB Oatmeal | PB<br>Oats (rolled)<br>2% milk<br>Cinnamon | 4<br>9.5 g<br>4<br>0 g<br>9<br>9 g<br>1.<br>3 g | Cinnamon Rice Pudding | Rice milk<br>Sugar<br>Rice (white)<br>Vanilla<br>Cinnamon | 2<br>43 g<br>4<br>8 g<br>4<br>5 g<br>2<br>g<br>0.<br>7 g |
| PB Cookies | PB<br>Egg<br>Sugar | 6<br>4 g<br>4<br>0 g<br>1<br>6 g | Blueberry Muffins | All-purpose flour<br>Sugar<br>Salt<br>Baking powder<br>Canola oil<br>Applesauce (unsweetened)<br>Rice milk<br>Vanilla<br>Blueberries (frozen) | 4<br>0 g<br>2<br>8 g<br>0.<br>5 g<br>1.<br>4 g<br>2<br>2 g<br>5<br>0 g<br>3<br>0 g<br>2<br>g<br>4<br>6 g |
| PB Sandwich | Whole wheat bread<br>PB<br>Jelly (strawberry or grape) | 4<br>8 g<br>5<br>5 g<br>2<br>3 g | Bread with Butter & Jam, and Grapes | White bread<br>Butter<br>Jam (strawberry)<br>Grapes | 4<br>8 g<br>2<br>8 g<br>4<br>0 g<br>7<br>6 g |
| PB Oat Balls | PB<br>Oats (rolled)<br>Honey | 6<br>7 g<br>2<br>2 g | Oat & Honey Balls | Butter<br>Oats (rolled)<br>Honey<br>Chocolate chips | 1<br>8 g<br>3<br>8 g |

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| | | | | | |
|---|---|---|---|---|---|
| | | 1<br>0.5 g | | Cranberries (dried) | 3<br>8 g<br>1<br>0 g<br>2<br>0 g |
| Asian-Inspired Trail Mix | Peanuts<br>Wasabi peas<br>Wheat Chex<br>Chickpeas (roasted)<br>Chinese 5 spice | 2<br>7 g<br>2<br>7 g<br>1<br>5 g<br>3<br>6 g<br>2<br>g | Guacamole & Chips | Avocado<br>Olive oil<br>Lime<br>Salt<br>Blue corn tortilla chips | 7<br>5 g<br>1<br>5 g<br>5<br>g<br>1.<br>5 g<br>5<br>0 g |
| Peanuts, Raisins, Spicy Cheese | Peanuts<br>Raisins<br>Pepper Jack cheese | 4<br>2 g<br>4<br>0 g<br>3<br>8 g | Cereal, Milk & Fruit Juice | Honey Ohs! cereal<br>Rice milk<br>Grape juice | 6<br>0 g<br>2<br>27 g<br>2<br>27 g |
| PB with Celery & Raisins | PB<br>Raisins<br>Celery | 7<br>5 g<br>1<br>7 g<br>1<br>10 g | Mashed Sweet Potatoes | Sweet potato<br>Butter | 2<br>55 g<br>3<br>7 g |
| Peanut Protein Pack | Peanuts<br>Cheddar cheese<br>Salami<br>Pepperoni<br>Crackers (butter) | 2<br>8 g<br>2<br>6 g<br>1<br>7 g<br>1<br>5 g<br>1<br>7 g | Dried Fruit & Crackers | Cranberries (dried)<br>Mango (dried)<br>Crackers (butter) | 6<br>0 g<br>3<br>4 g<br>4<br>0 g |

Each snack contained 500 +/- 10 kcal. Each PNT snack contained 20 +/- 1 g protein. Each CHO snack contained ~5 g protein. Peanuts were unsalted; peanut butter was labeled natural and contained no added ingredients other than salt. PNT: peanut, CHO: carbohydrate, PB: peanut butter.

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Supplemental Table 2. Post-intervention questionnaire results

| | PNT (N=14) | CHO (N=14) |
|---|---|---|
| **Exit survey** | | |
| Previously attempted to gain weight (%) | 64 | 21 |
| Mentally prepared to gain weight (%) | 86 | 79 |
| Food intake habits remained the same (%) | 86 | 50* |
| Eating 500 kcal on top of normal diet was difficult (%) | 21 | 29 |
| Sleep habits were fairly consistent throughout the study (%) | 86 | 57 |
| Exercise habits were fairly consistent throughout the study (%) | 64** | 93 |
| | **PNT (N=16)** | **CHO (N=15)** |
| **Snack acceptability†** | | |
| Easy to consume (%) | 94 | 93 |
| Acceptable (%) | 100 | 80 |
| Tasted great (%) | 94 | 80 |
| Snack choices represented a good variety (%) | 94 | 93 |
| Stayed fresh (%) | 100 | 93 |
| Easy to add to normal daily diet (%) | 88 | 73 |

PNT: peanut, CHO: carbohydrate. *Some individuals in the CHO group reported that they believed their food intake had increased overall. **Some individuals in the PNT group reported they were not able to continue their habitual aerobic exercise during the intervention. †The snack questionnaire asked the participants to rank the statements using a Likert scale (1= strongly disagree and 5= strongly agree).

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Supplemental Table 3. Linear mixed effects model results for examining change in body composition, dietary intake, and physical activity over time by sex of participants (N=32)

| Outcome | **Sex** (Male vs. Female) | | **Time** (Weeks) | | **Sex x Time** | |
|---|---|---|---|---|---|---|
| | **Estimate (SE)** | **-value** | **Estimate (SE)** | **-value** | **Estimate (SE)** | **-value** |
| **Body composition** | | | | | | |
| TBM (kg) | 14.47 (3.08) | 0.001 | 0.22 (0.03) | 0.001 | -0.04 (0.04) | .310 |
| LBM (kg) | 16.31 (2.36) | 0.001 | 0.16 (0.02) | 0.001 | -0.03 (0.03) | .294 |
| FM (kg) | -2.59 (1.49) | .086 | 0.06 (0.01) | 0.001 | -0.003 (0.02) | .877 |
| Body fat (%) | -9.21 (1.70) | 0.001 | -0.004 (0.02) | .277 | 0.04 (0.03) | .159 |
| **Dietary intake** | | | | | | |
| Total calories (kcal/d) | 662.40 (231.88) | .005 | 24.02 (15.05) | .031 | -4.71 (19.77) | .812 |
| CHO (g/kg/d) | 58.37 (32.87) | .079 | 2.89 (1.86) | .028 | -0.34 (2.44) | .889 |
| PRO (g/kg/d) | 28.38 (9.80) | .005 | 0.53 (0.76) | .086 | 0.67 (1.00) | .504 |
| Total fat (g/d) | 34.21 (11.00) | .003 | 1.14 (0.89) | .319 | -1.11 (1.17) | .343 |
| **Physical activity** | | | | | | |
| Daily steps | -1556.05 (1190.73) | .195 | -79.61 (87.90) | .259 | 27.91 (115.48) | .810 |

Significant difference $P<0.05$. PNT: peanut, CHO: carbohydrate, SE: standard error, TBM: total body mass, LBM: lean body mass, FM: fat mass, d: day, PRO: protein.

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Supplemental Table 4. BSL to post-intervention results for circumferences and health outcomes of participants assigned to the PNT and CHO snack groups (N=32)

| | **Group** | **BSL** | **Post** | **Estimate (SE)** | ***P*-value** |
|---|---|---|---|---|---|
| **Circumferences ‡** (cm) | | | | | |
| Waist | PNT | 80.0±7.6 | 80.9±8.4 | 0.75 (0.64) | 0.251 |
| | CHO | 79.9±5.8 | 81.9±6.0 | | |
| | PNT+CHO | 80.0±6.7 | 81.4±7.2 | | |
| Hip | PNT | 95.6±5.2 | 96.1±4.5 | 0.84 (0.49) | 0.094 |
| | CHO | 97.6±5.7 | 99.4±4.9 | | |
| | PNT+CHO | 96.6±5.4 | 97.7±4.9 | | |
| Calf | PNT | 35.6±2.6 | 35.7±2.6 | 0.10 (0.14) | 0.490 |
| | CHO | 37.1±2.4 | 37.5±2.3 | | |
| | PNT+CHO | 36.4±2.6 | 36.6±2.6 | | |
| Mid-arm | PNT | 29.7±4.2 | 30.6±4.0 | 0.04 (0.33) | 0.910 |
| | CHO | 30.7±3.8 | 31.6±3.9 | | |
| | PNT+CHO | 30.2±4.0 | 31.1±3.9 | | |
| Relaxed arm | PNT | 30.4±4.2 | 31.2±3.8 | 0.28 (0.289) | 0.339 |
| | CHO | 31.3±3.9 | 32.5±4.0 | | |
| | PNT+CHO | 30.9±4.0 | 31.9±3.9 | | |
| Flexed arm | PNT | 31.3±4.6 | 32.3±4.2 | 0.04 (0.33) | 0.898 |
| | CHO | 32.1±4.0 | 33.2±4.0 | | |
| | PNT+CHO | 31.7±4.3 | 32.7±4.1 | | |
| **Serum lipids†** (mg/dL) | | | | | |
| Total cholesterol | PNT | 163 (132-234) | 160 (134-236) | -1.56 (6.39) | 0.808 |
| | CHO | 147 (125-174) | 148 (110-213) | | |
| | PNT+CHO | 158 (125-234) | 154 (110-236) | | |
| HDL | PNT | 48 (39-78) | 47 (39-80) | -1.06 (2.68) | 0.695 |
| | CHO | 51 (35-72) | 51 (35-71) | | |
| | PNT | 50 | 50 (35- | | |

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| | | | | | |
|---|---|---|---|---|---|
| | T+CHO | (35-78) | 80) | | |
| LDL | PNT<br>CHO<br>PNT+CHO | 95 (61-161)<br>83 (58-110)<br>89 (58-161) | 96 (66-176)<br>84 (43-140)<br>90 (43-176) | -0.13 (6.24) | 0.984 |
| Triglycerides | PNT<br>CHO<br>PNT+CHO | 77 (42-194)<br>64 (49-214)<br>67 (42-214) | 72 (52-255)<br>69 (47-165)<br>72 (47-255) | -0.81 (9.74) | 0.934 |
| **Fasting blood glucose**† (mg/dL) | PNT<br>CHO<br>PNT+CHO | 92 (78-109)<br>89 (82-99)<br>90 (78-109) | 91 (85-112)<br>89 (79-104)<br>90 (79-112) | 0.25 (1.92) | 0.897 |
| **Blood pressure** ‡ (mmHg) | | | | | |
| Systolic | PNT<br>CHO<br>PNT+CHO | 111±11<br>112±10<br>112±10 | 113±11<br>112±12<br>112±11 | 1.23 (2.79) | 0.663 |
| Diastolic | PNT<br>CHO<br>PNT+CHO | 63±8<br>60±6<br>62±7 | 63±6<br>60±5<br>62±6 | 0.64 (1.59) | 0.691 |
| **Resting heart rate** ‡ (bpm) | PNT<br>CHO<br>PNT+CHO | 60±7<br>61±8<br>60±7 | 67±10<br>63±10<br>65±10 | 5.43 (2.85) | 0.066 |

Data reported as mean±SD. ‡N=31. †Data reported as median (range). *P*-values are based on the effect of snack group from a general linear model. BSL: baseline, PNT: peanut, CHO: carbohydrate, HDL: high density lipoprotein, LDL: low density lipoprotein, bpm: beats per minute.

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